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Chapter 1: State-space modelling and Kalman filter equations

The aim of this homework is to estimate the position of a football during a penalty kick when only noisy position measurements are available. The initial speed is not known exactly, therefore the estimator must combine the model prediction with the sensor information. A Kalman filter is appropriate for this problem because the adopted model is linear and the uncertainties are represented through covariance matrices.

The motion is described in a two-dimensional plane. The x-axis is directed from the penalty spot to the goal line, while the y-axis describes the lateral displacement of the ball. The ball starts from the origin and reaches the goal when x = L. The goal is considered successful if |y| remains smaller than the half-width W at x = L.

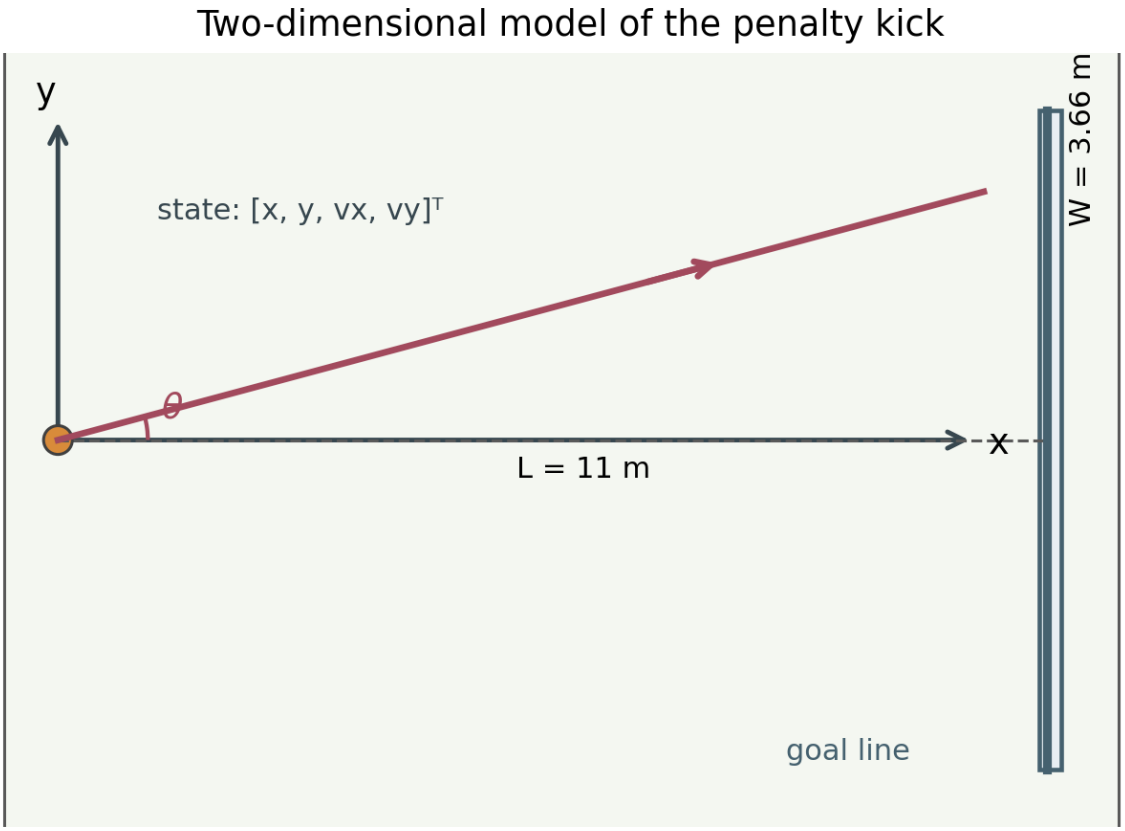


Figure 1. Two-dimensional geometry used for the penalty-kick model.

1.1 Continuous-time model

A constant-velocity model is used. Small unmodelled effects, such as aerodynamic deviations and modelling approximations, are included as process noise acting on the velocity derivatives. The state vector is

$$\mathbf{X} = [\, x \quad y \quad vx \quad vy \,]^T$$

with the continuous-time equations

$$\dot{x} = vx, \quad \dot{y} = vy, \quad \dot{v}x = w1x, \quad \dot{v}y = w1y$$

The measured output is the noisy position of the ball:

$$yobs = [\, x \quad y \,]^T + w2.$$

The corresponding matrices are

$$A = [[0,0,1,0],[0,0,0,1],[0,0,0,0],[0,0,0,0]], \quad F = [[0,0],[0,0],[1,0],[0,1]], \quad C = [[1,0,0,0],[0,1,0,0]].$$

Because the outputs are x and y, the pair (A, C) is observable. The observability matrix has rank 4, equal to the number of states, so the four states can be reconstructed from the measured positions over time.

1.2 Numerical assumptions

The geometric data are selected from the standard football penalty setup. The distance from the penalty mark to the goal line is L = 11 m and the goal width is 7.32 m, therefore W = 3.66 m. The admissible shooting angle from the penalty mark is approximately atan(W/L) = 18.40 deg. A 20% margin is added to account for modelling uncertainty, giving an angular bound of about 22.08 deg.

Quantity	Value	Meaning
L	11 m	distance from the penalty spot to the goal line
W	3.66 m	half of the goal width
θmax	18.40 deg	geometric maximum angle for the goal mouth
θlim	22.08 deg	angle bound with 20% margin
vx range	[23.17, 35] m/s	forward velocity component range
vy range	[-13.16, 13.16] m/s	lateral velocity component range
$\bar{x}0$	$[0, 0, 29.25, 0]^T$	mean initial state
P0	diag(0, 0, 8.33, 43.79)	initial covariance
Q	diag(28.3, 28.3)	continuous acceleration-noise covariance
R	diag(0.00694, 0.00694)	position measurement-noise covariance

These values keep the simulated velocity uncertainty consistent with the random speed and direction of the kick, while the measurement covariance corresponds to a realistic noisy position sensor.

1.3 Discrete Kalman filter

Although the physical model is continuous, the sensor supplies data at discrete time instants. For this reason, the estimator is implemented in discrete time with a sampling time Ts = 0.01 s. The discrete model is

$$x_{k+1} = A_d x_k + w_{1k}, \qquad y_k = C x_k + w_{2k}$$

where Ad = exp(A Ts). For the constant-velocity system, this gives

$$A_d = [[1,0,T_s,0],[0,1,0,T_s],[0,0,1,0],[0,0,0,1]].$$

The discrete Kalman filter then alternates between prediction and correction. The prediction estimates where the ball should be according to the model, and the correction uses the measured position to reduce the estimation error.

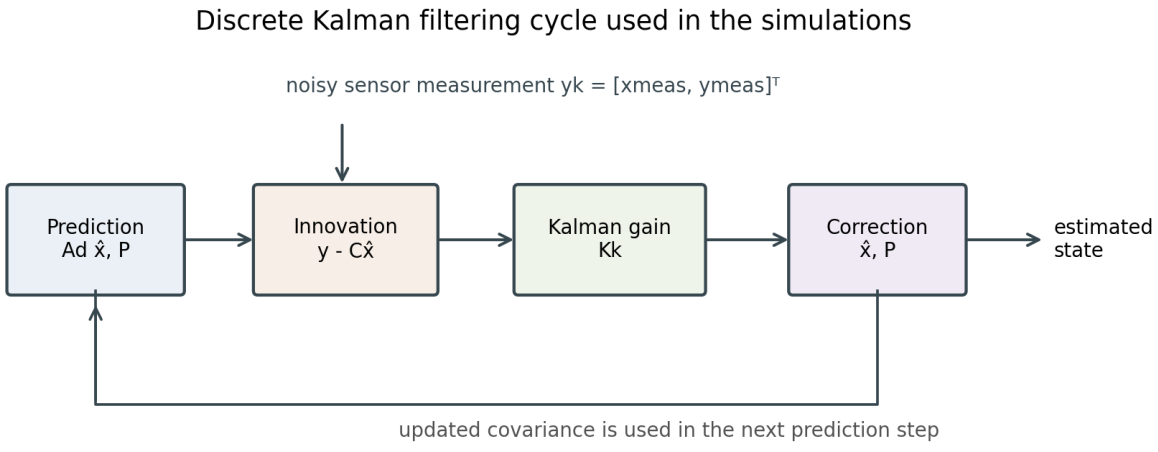


Figure 2. Prediction-correction structure of the discrete Kalman filter.

The position estimation error used for validation is the Euclidean distance between the estimated and true noiseless position:

$$e(k) = \sqrt{(\hat{x}(k) - x_{true}(k))^2 + (\hat{y}(k) - y_{true}(k))^2}.$$

Chapter 2: Random penalty-kick simulations

Ten penalty kicks were simulated using different initial speeds and shooting angles. The initial speed was selected in the interval 25-35 m/s and the angle was chosen inside the admissible angular range. The following table lists the test cases used in the numerical study.

Kick	Initial speed V0 [m/s]	Angle θ [deg]
1	25.835	-21.33
2	27.467	18.91
3	30.860	3.597
4	31.260	-16.74
5	31.609	16.02
6	31.664	3.537
7	32.298	-0.6935
8	32.386	11.88
9	33.908	15.23
10	34.823	-12.83

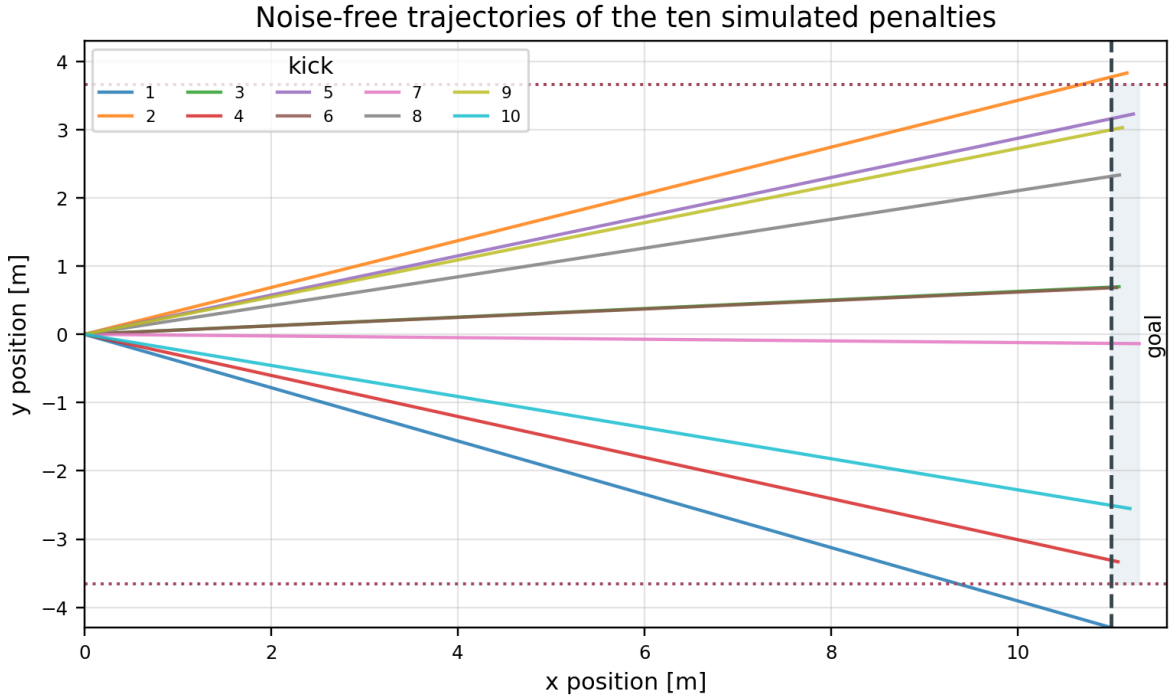


Figure 3. Noise-free trajectories for the ten simulated penalty kicks.

For each test case, the true trajectory is generated from the constant-velocity model. Gaussian measurement noise is added to x and y, and the Kalman filter uses these noisy measurements to reconstruct the state. The error is evaluated over the whole flight and especially at the instant when the ball reaches x = 11 m.

Kick	V0 [m/s]	θ [deg]	reach time [s]	error at goal [m]	< 0.25 m
1	25.835	-21.330	0.4571	0.0581	Yes
2	27.467	18.914	0.4233	0.0118	Yes
3	30.860	3.597	0.3572	0.0294	Yes
4	31.260	-16.743	0.3675	0.0153	Yes
5	31.609	16.017	0.3621	0.0142	Yes
6	31.664	3.537	0.3481	0.0101	Yes
7	32.298	-0.693	0.3406	0.0091	Yes
8	32.386	11.880	0.3471	0.0314	Yes
9	33.908	15.229	0.3362	0.0436	Yes
10	34.823	-12.833	0.3240	0.0204	Yes

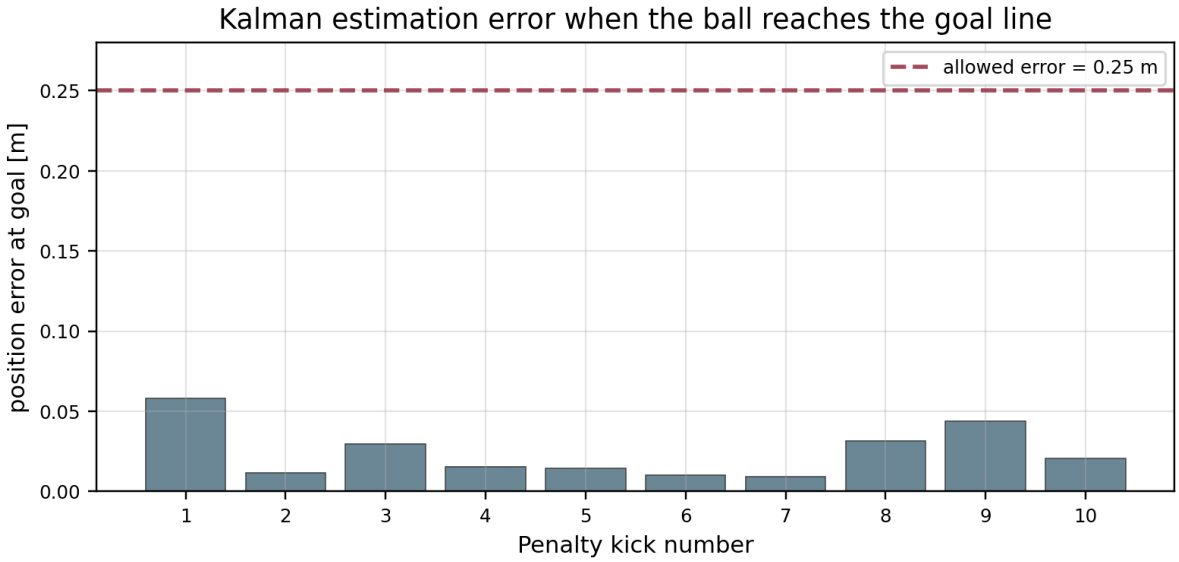


Figure 4. Estimation error at the goal line for all penalty kicks.

2.1 Individual simulation results

The following figures show the measured, true and estimated x-y position components for each kick. The final subplot gives the norm of the position error and compares it with the 0.25 m admissible threshold.

Kick 1

For this kick, V0 = 25.835 m/s and θ = -21.330 deg. The ball reaches the goal line at t = 0.4571 s, and the estimated position error at that instant is 0.0581 m.

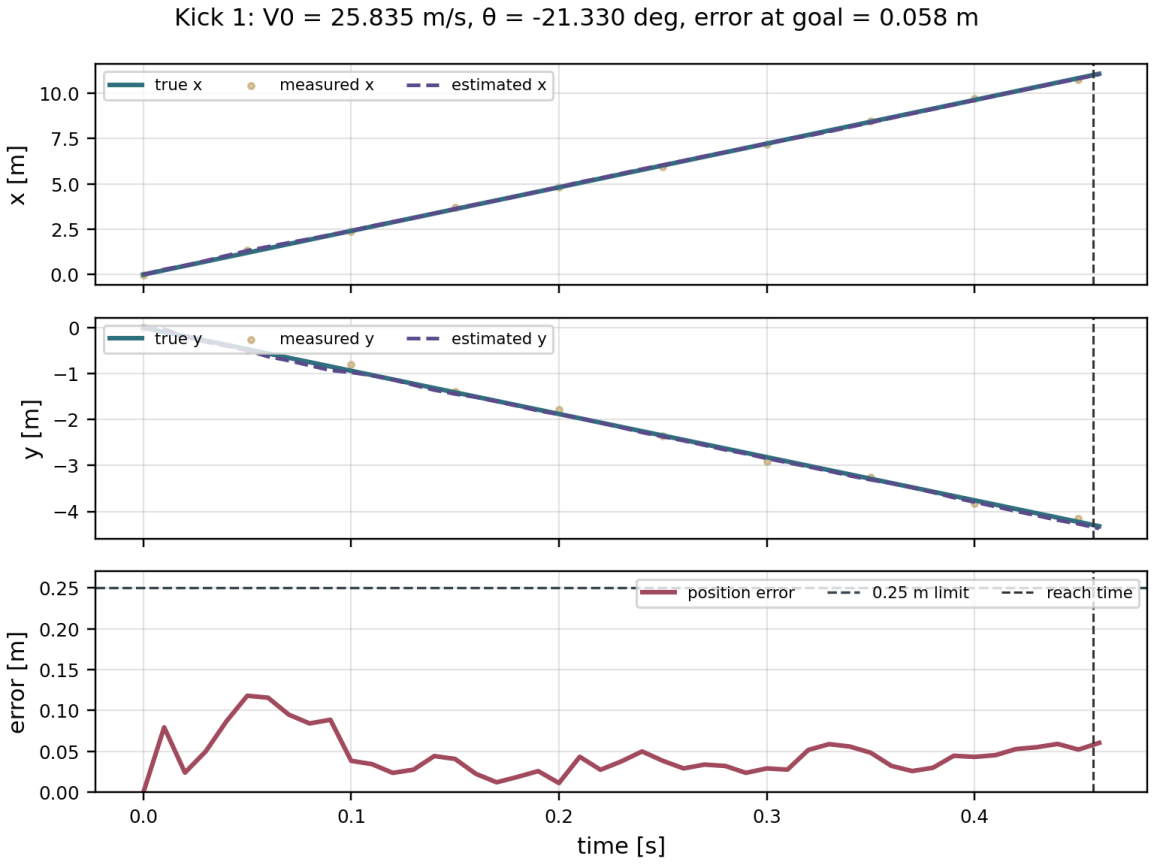


Figure 5. True, measured and estimated positions, with estimation error, for kick 1.

Kick 2

For this kick, V0 = 27.467 m/s and θ = 18.914 deg. The ball reaches the goal line at t = 0.4233 s, and the estimated position error at that instant is 0.0118 m.

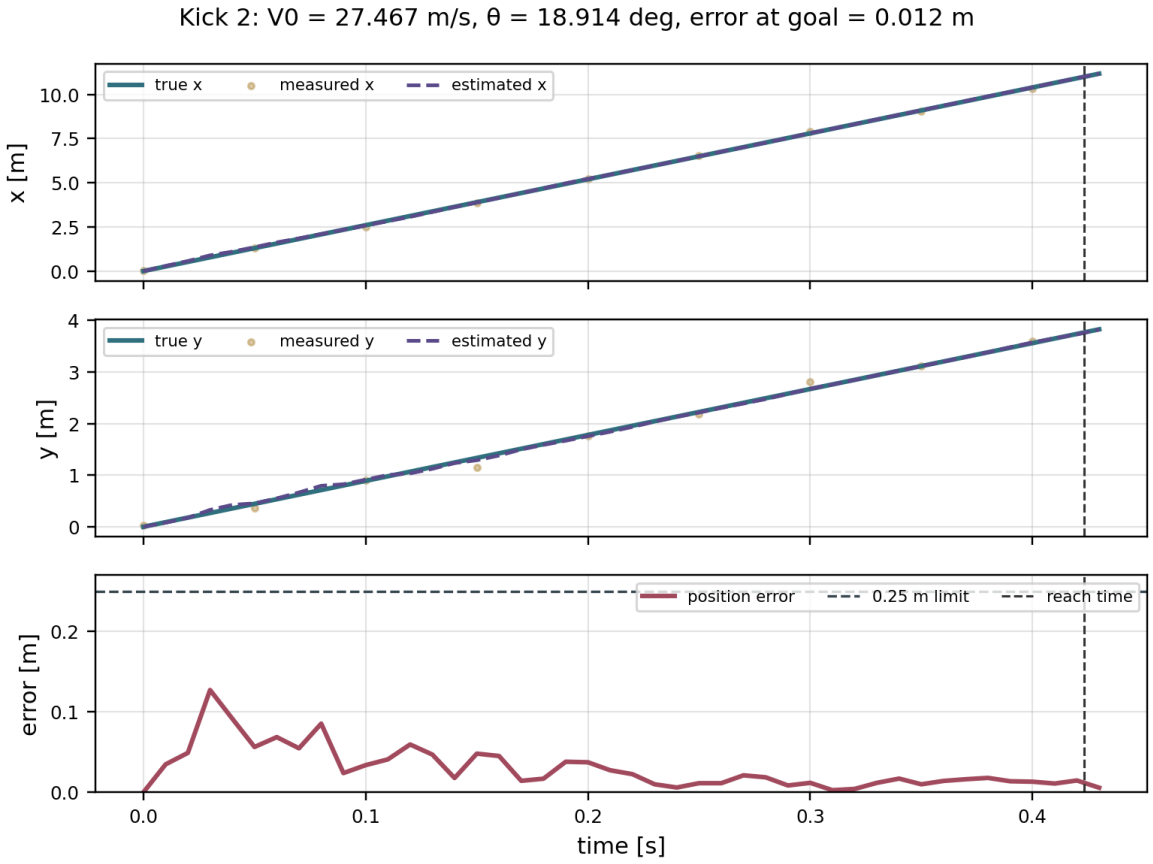


Figure 6. True, measured and estimated positions, with estimation error, for kick 2.

Kick 3

For this kick, V0 = 30.860 m/s and θ = 3.597 deg. The ball reaches the goal line at t = 0.3572 s, and the estimated position error at that instant is 0.0294 m.

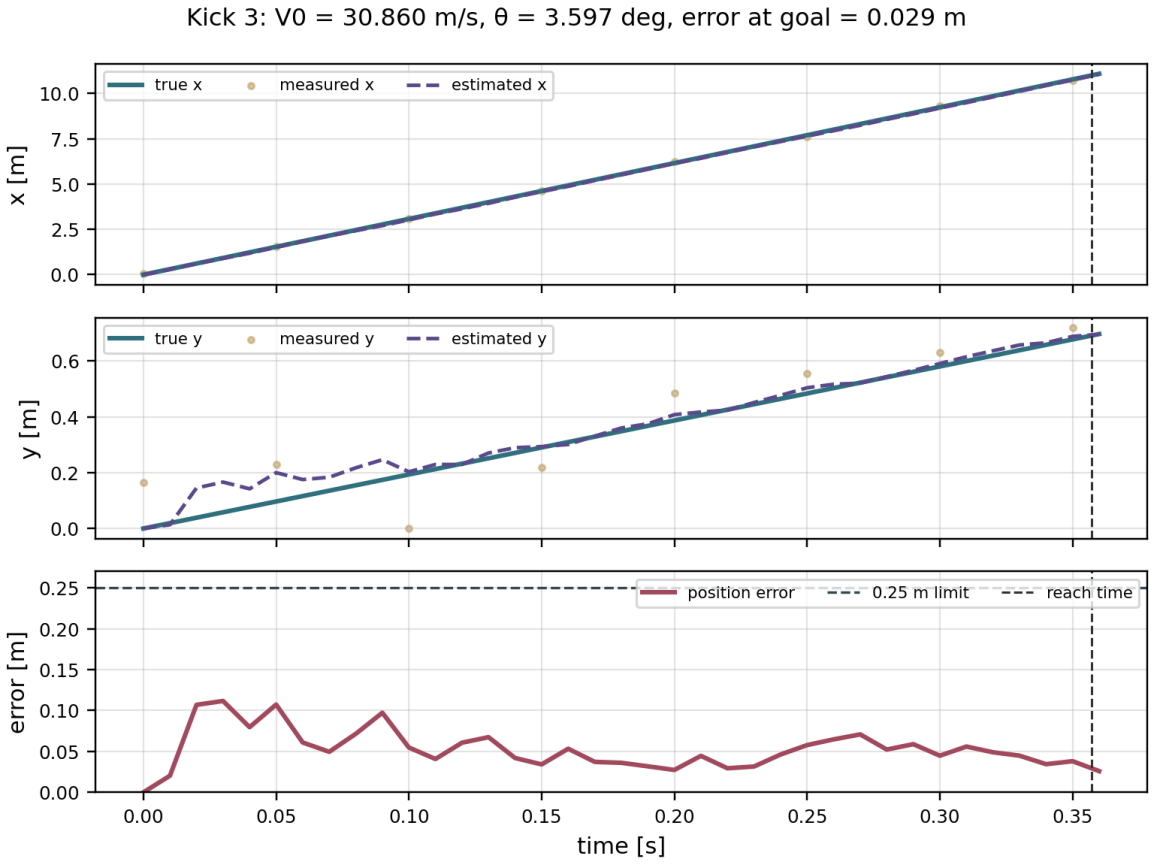


Figure 7. True, measured and estimated positions, with estimation error, for kick 3.

Kick 4

For this kick, V0 = 31.260 m/s and θ = -16.743 deg. The ball reaches the goal line at t = 0.3675 s, and the estimated position error at that instant is 0.0153 m.

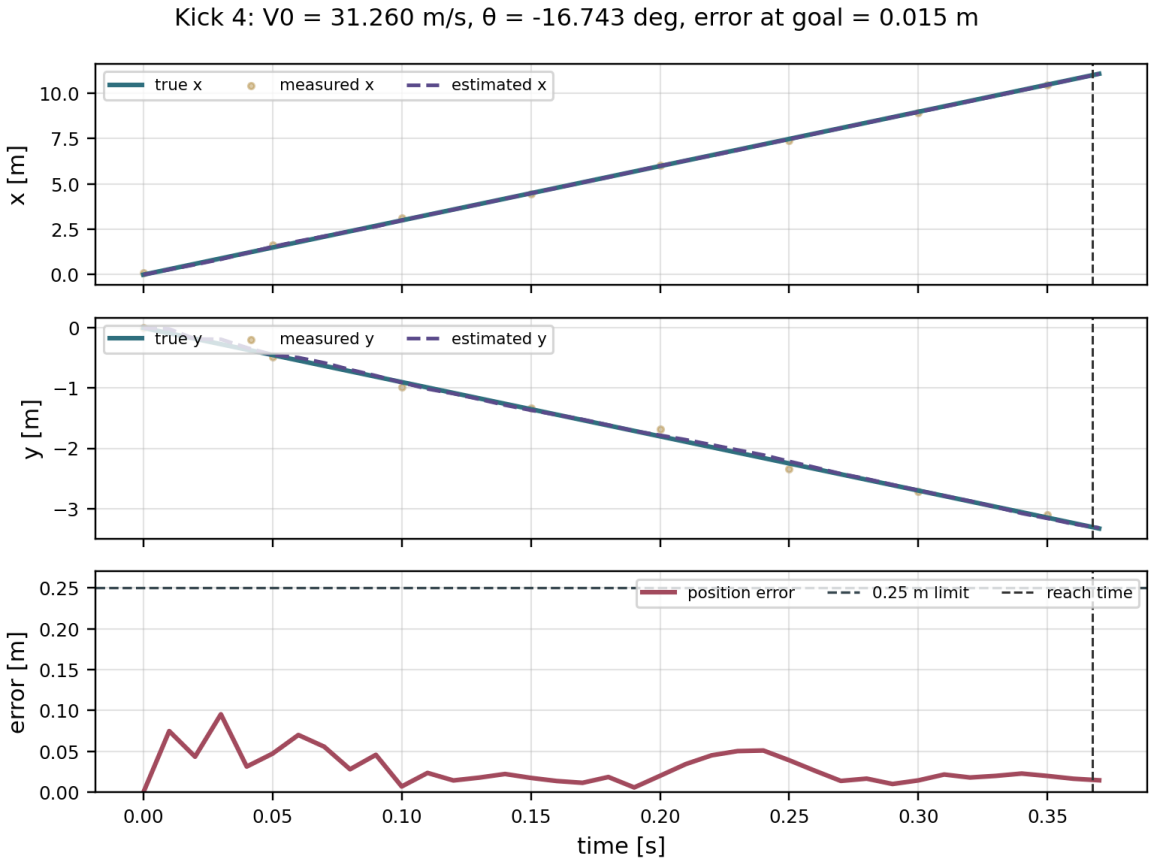


Figure 8. True, measured and estimated positions, with estimation error, for kick 4.

Kick 5

For this kick, V0 = 31.609 m/s and θ = 16.017 deg. The ball reaches the goal line at t = 0.3621 s, and the estimated position error at that instant is 0.0142 m.

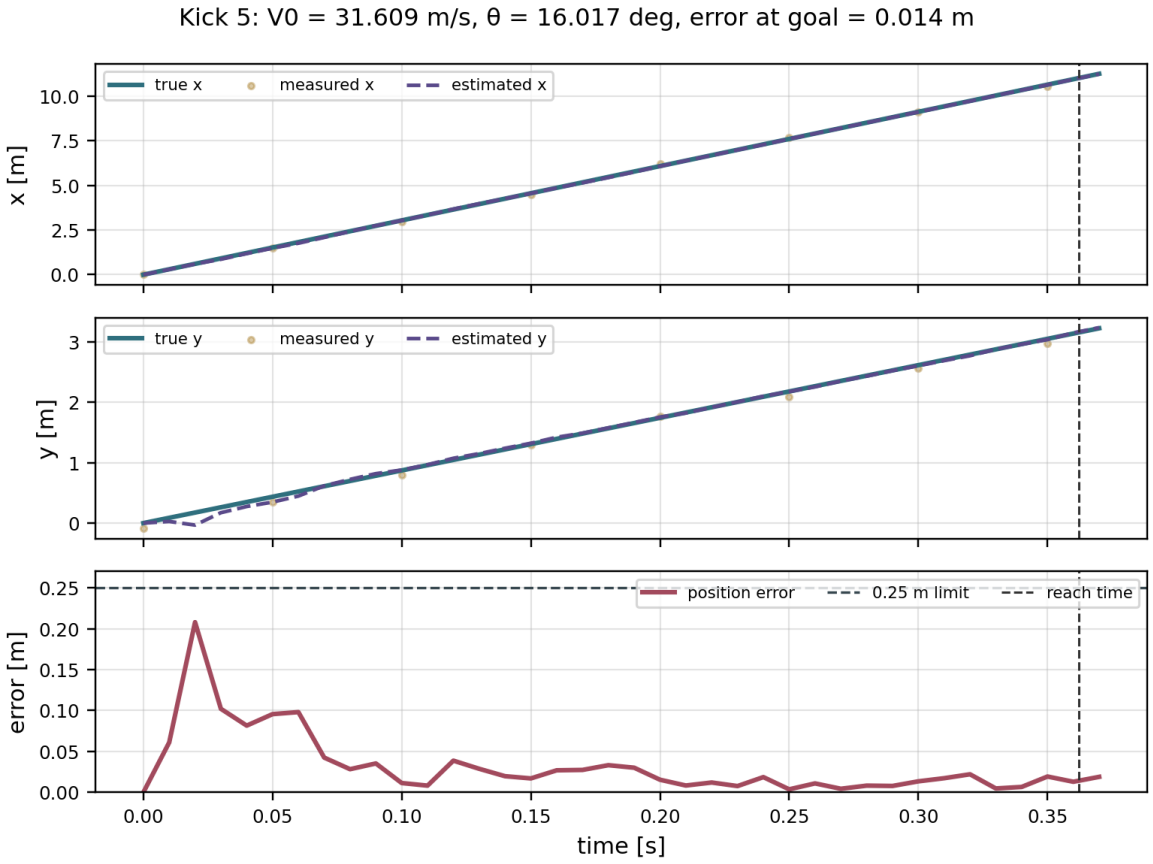


Figure 9. True, measured and estimated positions, with estimation error, for kick 5.

Kick 6

For this kick, V0 = 31.664 m/s and θ = 3.537 deg. The ball reaches the goal line at t = 0.3481 s, and the estimated position error at that instant is 0.0101 m.

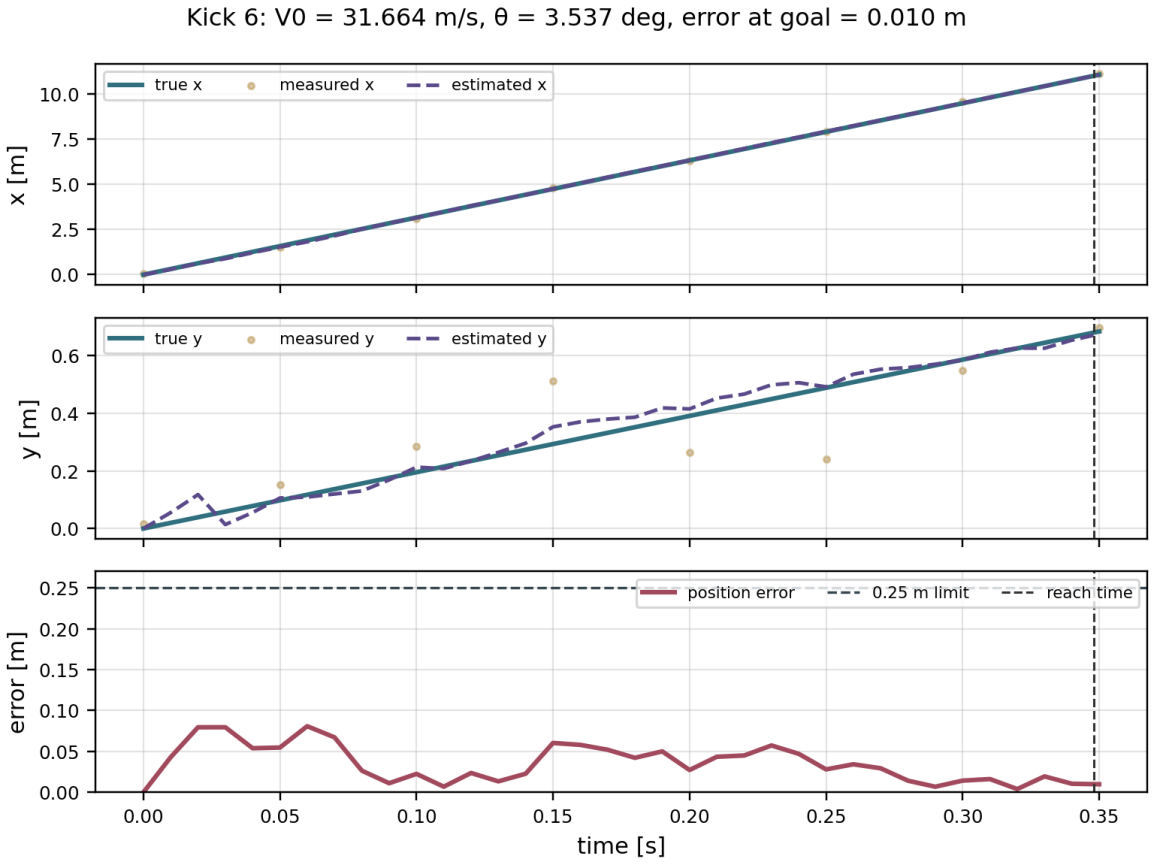


Figure 10. True, measured and estimated positions, with estimation error, for kick 6.

Kick 7

For this kick, V0 = 32.298 m/s and θ = -0.693 deg. The ball reaches the goal line at t = 0.3406 s, and the estimated position error at that instant is 0.0091 m.

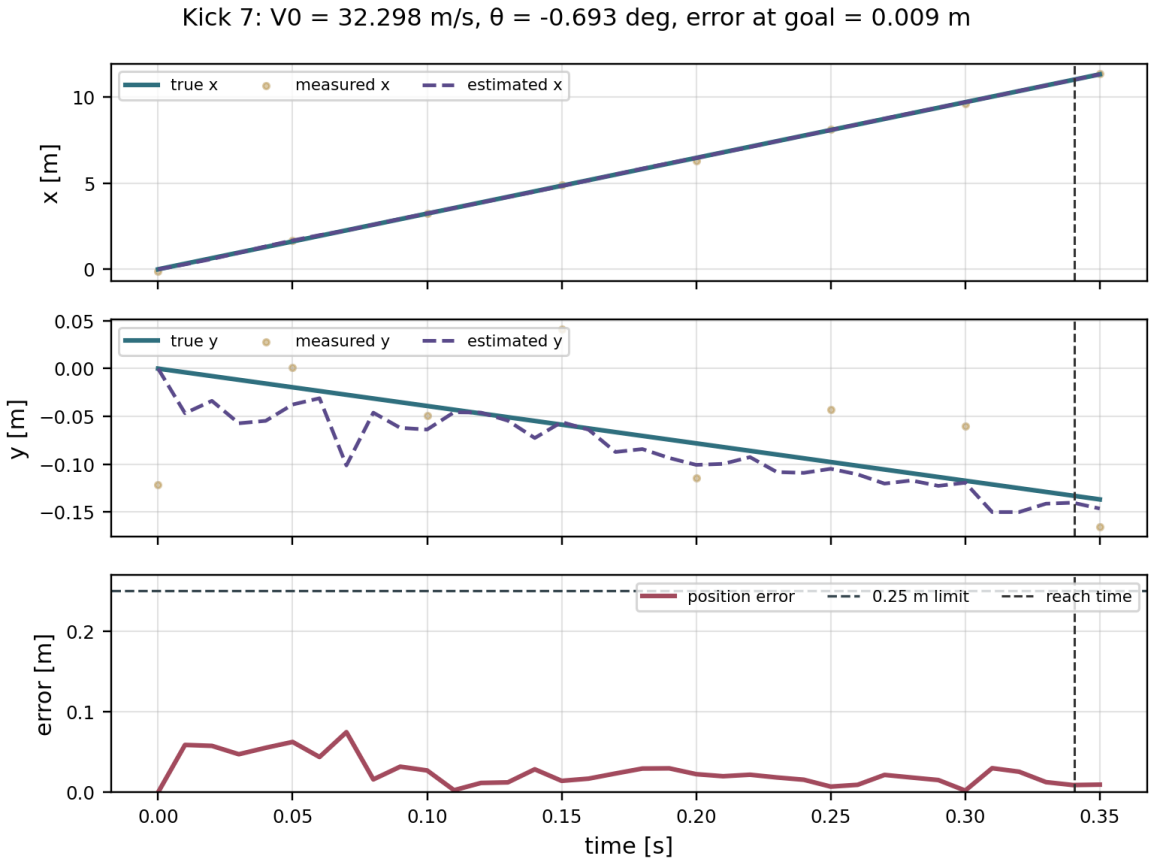


Figure 11. True, measured and estimated positions, with estimation error, for kick 7.

Kick 8

For this kick, V0 = 32.386 m/s and θ = 11.880 deg. The ball reaches the goal line at t = 0.3471 s, and the estimated position error at that instant is 0.0314 m.

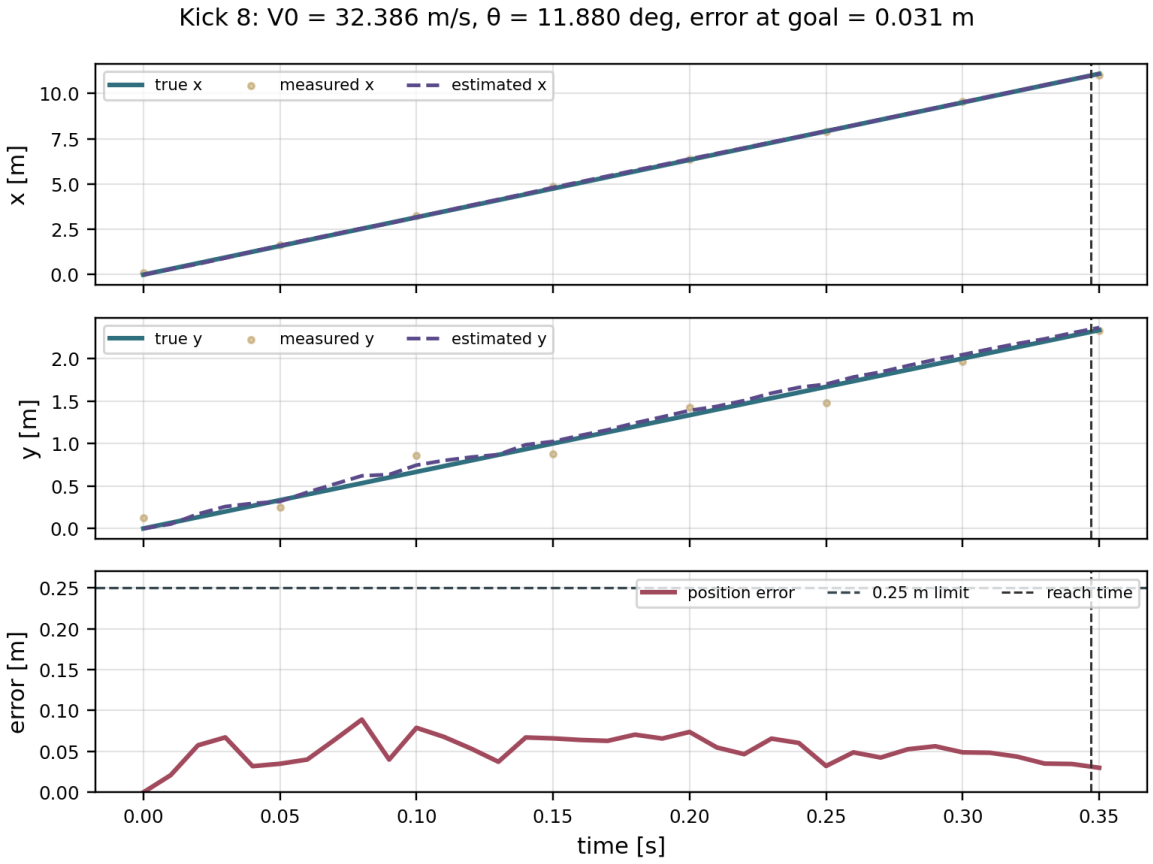


Figure 12. True, measured and estimated positions, with estimation error, for kick 8.

Kick 9

For this kick, V0 = 33.908 m/s and θ = 15.229 deg. The ball reaches the goal line at t = 0.3362 s, and the estimated position error at that instant is 0.0436 m.

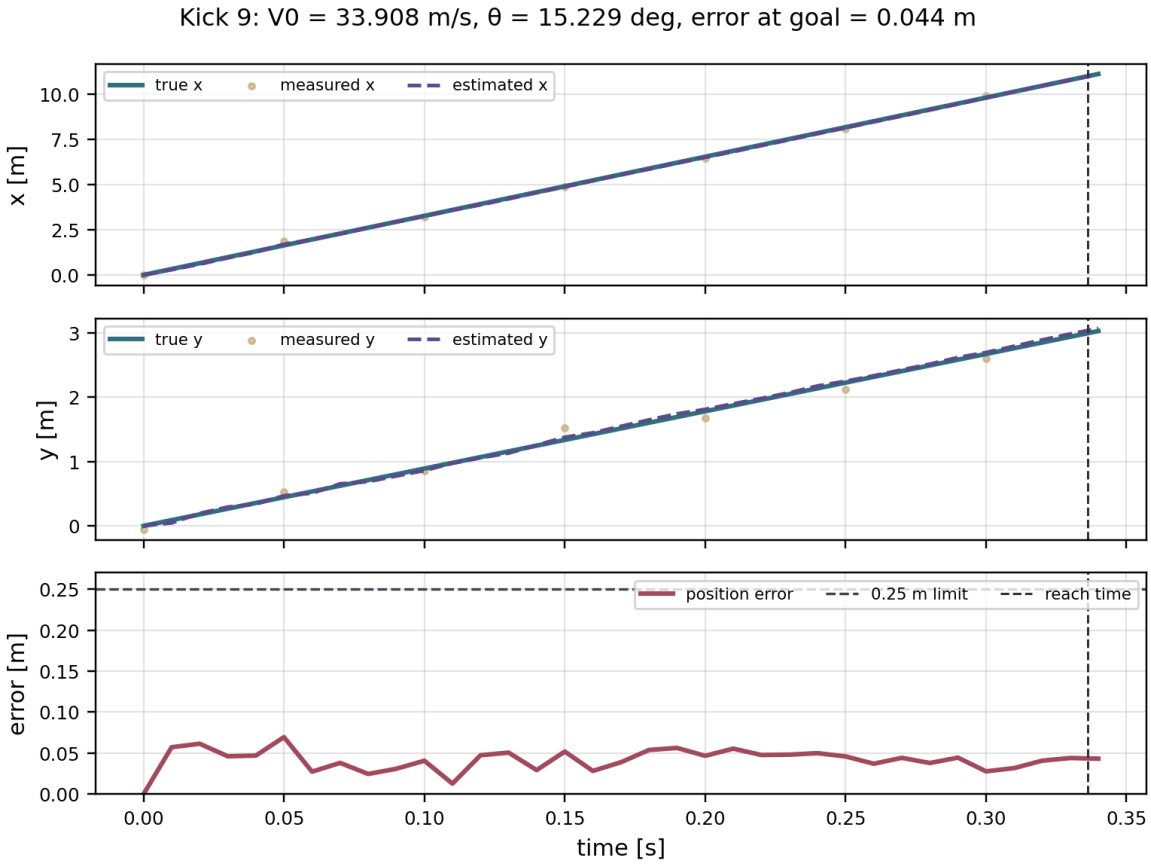


Figure 13. True, measured and estimated positions, with estimation error, for kick 9.

Kick 10

For this kick, V0 = 34.823 m/s and θ = -12.833 deg. The ball reaches the goal line at t = 0.3240 s, and the estimated position error at that instant is 0.0204 m.

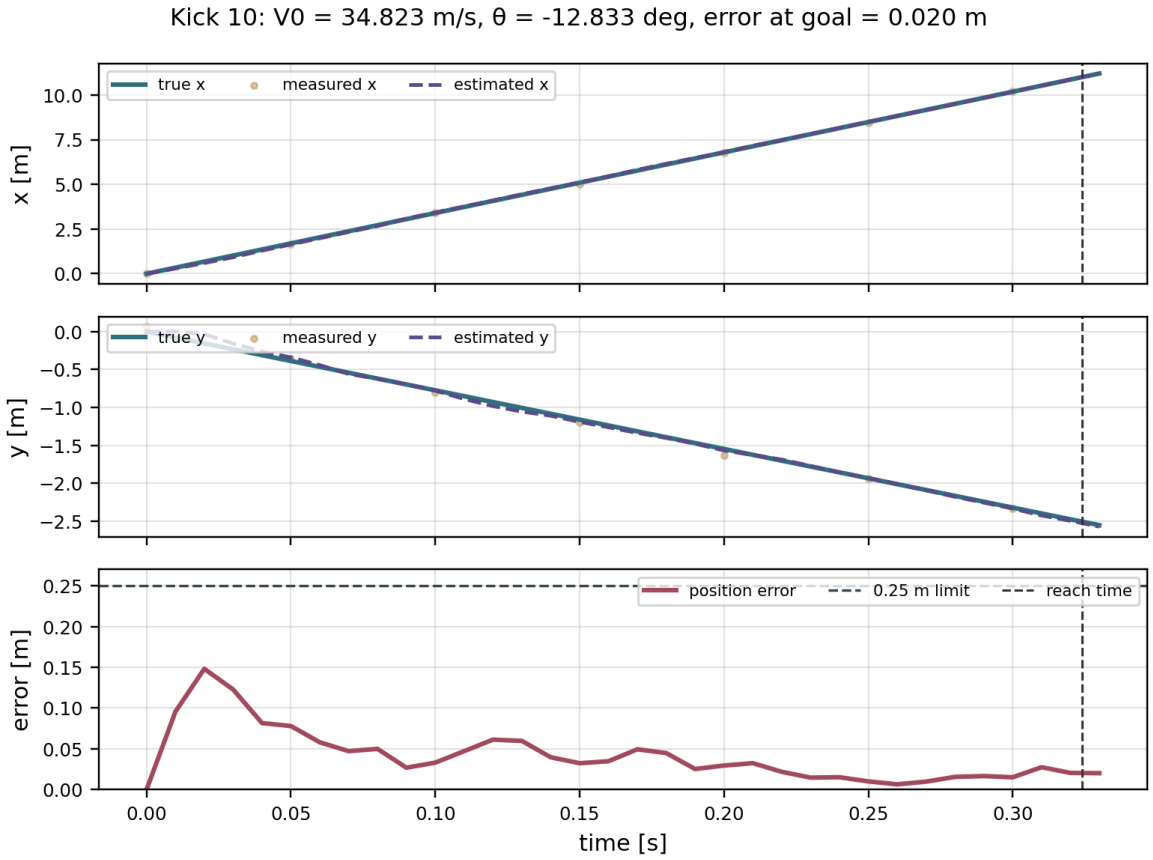


Figure 14. True, measured and estimated positions, with estimation error, for kick 10.

Chapter 3: Discussion of numerical results

All simulated kicks satisfy the required accuracy condition at the goal line. The largest error at reach time is 0.0581 m, which is still well below the admissible limit of 0.25 m. The mean error over the ten simulations is 0.0243 m. The smallest final error occurs in kick 7, while the largest one occurs in kick 1.

The plots also show the expected behaviour of the Kalman filter. At the beginning of the shot, the velocity is uncertain, so the model prediction can be imperfect. After several measurements, the estimator corrects the state and the position error remains bounded. The filter is able to smooth the noisy measurements without delaying the true trajectory significantly.

Conclusion

A discrete-time Kalman filter was designed for the estimation of a football position during a penalty kick. Starting from a simple constant-velocity model, the system was written in state-space form and observability was verified using position measurements in both coordinates. The numerical parameters were selected from the geometry of a real penalty kick and from the required measurement accuracy.

The estimator was tested on ten different shots with different speeds and directions. In all cases the estimation error at the goal line remained below 0.25 m, confirming that the filter provides accurate position estimates for the considered measurement noise and model uncertainty. The resulting estimator is therefore suitable for tracking the ball position during the short flight time of a penalty kick.